Chapter 3
Vector Spaces

3.6 Row Space and Column Space

Row space & column space

- For an $m \times n$ matrix A, the row vectors are n-vectors from $\mathbb{R}^{1\times n}$, and the column vectors are m-vectors from $\mathbb{R}^m$.

Definition

For an $m \times n$ matrix A, the subspace of $\mathbb{R}^{1\times n}$ spanned by the row vectors of A is called the **row space** of A.

The subspace of $\mathbb{R}^m$ spanned by the column vectors of A is called the **column space** of A.

$$span(\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}) = \{\boldsymbol{u} | \ \boldsymbol{u} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n\}$$

If $\mathbf{v}_i \in \mathbb{R}^m$, then $span(\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n})$ is a subspace of $\mathbb{R}^m$

Example

The 3×4 matrix

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 0 & -4 & 3 \\ 1 & -3 & -14 & 9 \end{bmatrix}$$

Has a row space in $\mathbb{R}^{1\times 4}$ given by

$$Span([1,1,2,-1],[2,0,-4,3],[1,-3,-14,9])$$

And the column space in $\mathbb{R}^3$ given by

$$Span\left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ -14 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 9 \end{bmatrix}\right)$$

Question: can we express row/column spaces as a span of fewer vectors?

Row space & Row Equivalence

Theorem

Two row-equivalent matrices have the same row space.

Proof

- If B is row equivalent to A, then B can be formed from A by a finite sequence of row operations. Thus the row vectors of B must be linear combinations of the row vectors of A.
- Consequently, the row space of B must be subspaces of the row space A.
- By the same reasoning, the row space of A must be subspaces of the row space of B.

Null space, column space from rref

Theorem

Two row-equivalent matrices have the same null space.

This is because null space is the solution set of Ax=0, and row operations does not change the solution set. This also means that, if U is the reduced row echelon form (rref) of A, then any linear relationship between column vectors of U, also holds between column vectors of A!

If U is the rref of A and $c_1\boldsymbol{u_1} + c_2\boldsymbol{u_2} + \cdots = \boldsymbol{0}$, then also $c_1\boldsymbol{a_1} + c_2\boldsymbol{a_2} + \cdots = \boldsymbol{0}$.

The columns of U with leading 1's form a basis of the column space of U, because they are linearly independent, and all other columns (corresponding to free variables) are in their span. So we also have the same relationship between columns of A!

All the column vectors of A corresponding to the leading variables in U form a basis of the column space of A.

Finding basis of Row, Column, Null space

Using the 2 previous slides, we can find a basis for the row space, column space, and null space of a matrix A, by row reduction:

1. Row reduce A into reduced row echelon form, U.
2. The non-zero row vectors of U forms a basis for row space of A.
3. The column vectors **of A** (not U!) corresponding to leading 1's in U forms a basis of column space of A
4. Read the equation U**x**=**0** row by row to find N(A); express it as a span of vectors.

Example

Find a basis for the row space of A, the column space of A and the null space of A, for $A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 5 & 4 & 3 \\ 2 & 0 & -2 & 1 \end{bmatrix}$.

Example

Find a basis for the row space of A, the column space of A and the null space of A, for $A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 5 & 4 & 3 \\ 2 & 0 & -2 & 1 \end{bmatrix}$.

Solution

- The reduced echelon form of A is given by $U = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- Row space: $\{[1,0,-1,0],[0,1,1,0],[0,0,0,1]\}$ is a basis for row space of A.
- Column space: The leading 1's of U occur in the first, second and fourth columns. Thus

$\left\{\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}\right\}$ form a basis for column space of A.

See next page for N(A)

Example (continued)

- Null space:

$$U\boldsymbol{x} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_1 - x_3 = 0, x_2 + x_3 = 0, x_4 = 0$$

$$\Rightarrow N(A) = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \\ 0 \end{bmatrix} : x_3 \in R \right\} = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right)$$

So a basis for $N(A)$ is $\{[1, -1, 1, 0]^T\}$.

Textbook Example 5

Find the dimension of the subspace of $\mathbf{R}^4$ spanned by

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 2 \\ 5 \\ -3 \\ 2 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 2 \\ 4 \\ -2 \\ 0 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 3 \\ 8 \\ -5 \\ 4 \end{pmatrix}$$

Textbook Example 5 (continued)

Solution

The subspace Span($\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, $\mathbf{x}_4$) is the same as the column space of the matrix

$$X = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 2 & 5 & 4 & 8 \\ -1 & -3 & -2 & -5 \\ 0 & 2 & 0 & 4 \end{bmatrix}$$

The row echelon form of X is

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The first two columns $\mathbf{x}_1$ and $\mathbf{x}_2$ of X will form a basis for the column space of X. Thus, dim Span($\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, $\mathbf{x}_4$) = 2.

Rank

Definition

The **rank** of a matrix A is the dimension of the row space of A, denoted $\text{rank}(A)$.

- Row equivalent matrices have the same row space,
- therefore, the Rank of A is equal to the rank of its reduced row echelon form.
- The rank of a reduced row echelon form is equal to the number of lead variables (i.e. the number of non-zero rows in rref)

Example

- Consider the matrix

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 1 \\ 1 & -4 & 7 \end{bmatrix}$$

- The row echelon form of A is:

$$U = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

- Clearly, $[1, -2,3]$ and $[0,1,5]$ form a basis for the row space of U. Since A and U are row equivalent, they have the same row space. Therefore, the rank of A is 2.

Dimension Theorem

Theorem

dim(Row Space A) = dim(Column Space A) = $\text{rank}(A)$.

Proof idea

The dimension of the column space of A is the number of linearly independent columns of A, that is the number of columns with a lead 1 in row echelon form of A, denoted U.

But this number is also the number of nonzero rows in U, that is the dimension of the row space of A.

Theorem

Consistency theorem for Linear Systems

The linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

Let A be an $m \times n$ matrix.

- The linear system $A\mathbf{x} = \mathbf{b}$ is **consistent** for every $\mathbf{b} \in \mathbb{R}^{\boldsymbol{m}}$ if and only if the column vectors of A **span** $\mathbb{R}^{\boldsymbol{m}}$.
- The system $A\mathbf{x} = \mathbf{b}$ has **at most one solution** for every $\mathbf{b} \in \mathbb{R}^{\boldsymbol{m}}$ if and only if the column vectors of A are **linearly independent**.

An $n \times n$ **square** matrix A is **nonsingular** if and only if the column vectors of A form a **basis** for $\mathbb{R}^{\boldsymbol{n}}$ ($\Leftrightarrow Ax = b$ always have unique solution $\Leftrightarrow rank(A) = n \Leftrightarrow \det(A) \neq 0$).

Nullity

Definition

The nullity of a matrix A is the dimension of the null space of A, that is nullity(A) = dim($N(A)$).

Theorem **rank(A)+dim(N(A)=n**

If A is an $m \times n$ matrix, then the rank of A plus the nullity of A equals n

The number of lead variables: rank(A)
The number of free variables: dim(N(A)
The number of variables: n

Rank, Nullity, and Solutions

Let $A \in \mathbb{R}^{m \times n}$. The following are equivalent:

- The columns of A are linearly independent
- $N(A) = \{\mathbf{0}\}$
- Nullity(A) = 0
- $A\boldsymbol{x} = \boldsymbol{b}$ has at most 1 solution for any $\boldsymbol{b}$

Let $A \in \mathbb{R}^{m \times n}$. The following are equivalent:

- The columns of A spans $\mathbb{R}^m$
- The column space of A is $\mathbb{R}^m$
- Rank(A) = m
- $A\boldsymbol{x} = \boldsymbol{b}$ always have at least 1 solution for any $\boldsymbol{b}$

True or False

a) If S is a subspace of a vector space V, then S is a vector space.

b) It is possible to find a pair of two-dimensional subspaces S and T of $\mathbb{R}^3$ such that $S \cap T = \{\mathbf{0}\}$.

c) If S and T are different proper subspaces of a vector space V, then $S \cup T$ is a subspace of V.

True or False

a) If S is a subspace of a vector space V, then S is a vector space.

b) It is possible to find a pair of two-dimensional subspaces S and T of $\mathbb{R}^3$ such that $S \cap T = \{\mathbf{0}\}$.

c) If S and T are different proper subspaces of a vector space V, then $S \cup T$ is a subspace of V.

Answer:

- a) True;
- b) False; two-dimensional subspaces are planes, but two planes cannot intersect only at **0**.
- c) False. Closure property C1 might not hold the union of two sets. For example, if S=$\{\begin{bmatrix} x \\ 0 \end{bmatrix} | x \in \mathbb{R}\}$ and T=$\{\begin{bmatrix} 0 \\ y \end{bmatrix} | y \in \mathbb{R}\}$, then although $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are both in $S \cup T$, their sum $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is in neither S nor T, so it is not in $S \cup T$, failing C1. $Note\ S \cup T$ is not $\mathbb{R}^2$

True or False

d) If S and T are subspaces of a vector space V, then $S \cap T$ is a subspace of V.

e) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ span $\mathbb{R}^n$, then they are linearly independent.

f) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ span a vector space V, then they are linearly independent.

True or False

d) If S and T are subspaces of a vector space V, then $S \cap T$ is a subspace of V.

e) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ span $\mathbb{R}^n$, then they are linearly independent.

f) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ span a vector space V, then they are linearly independent.

Answer:

- d) True. Closure of addition and scaler multiplication holds
- e) True. Rank(X)=n, and by theorem, nullity(X) = n-n=0, so they are linearly independent.
- f) False. (True if V has dimension =n)

True or False

h) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are vectors in vector space V and $\text{Span}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) = \text{Span}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{n-1})$ then $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are linearly dependent.

i) If A is an $m \times n$ matrix, then A and A^T have the same rank.

j) If A is an $m \times n$ matrix, then A and A^T have the same nullity.

True or False

h) If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are vectors in vector space V and $\text{Span}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) = \text{Span}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{n-1})$ then $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are linearly dependent.

i) If A is an $m \times n$ matrix, then A and A^T have the same rank.

j) If A is an $m \times n$ matrix, then A and A^T have the same nullity.

Answer:

h) True; By theorem 3.4.1 If {v1, v2, . . . , vn} is a spanning set for a vector space V, then any collection of m vectors in V, where m > n, is linearly dependent.

i) True; rank(A^T)=dim(Row Space A^{T}) = dim(Column Space A)=rank(A)

j) False; nullity (A)= n-rank(A)=n-rank(A^T) is not equal to m-rank(A^T) if m is not equal to n. Consider A=[1,2]. nullity (A)=1 but nullity(A^T) =0.